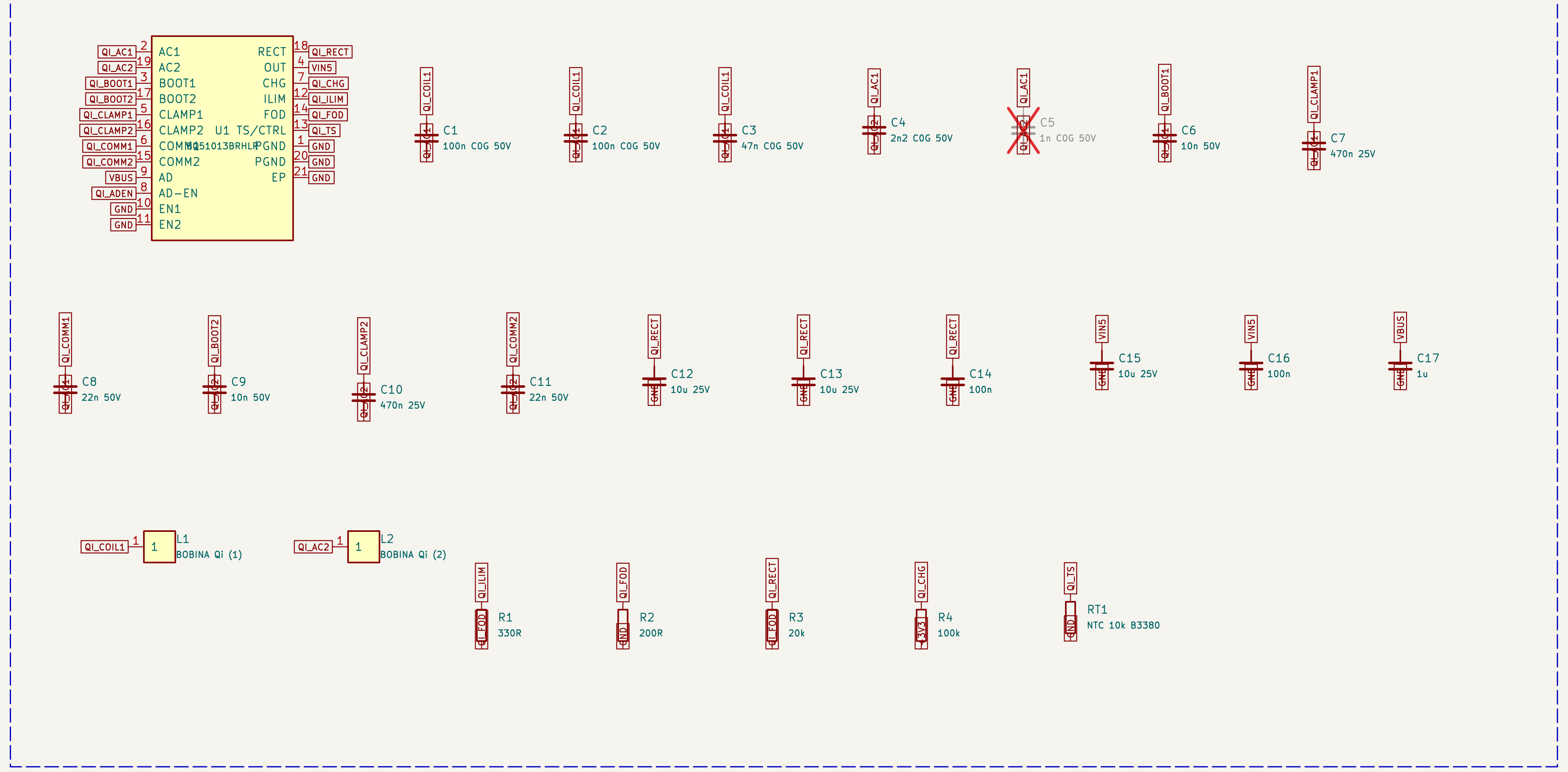
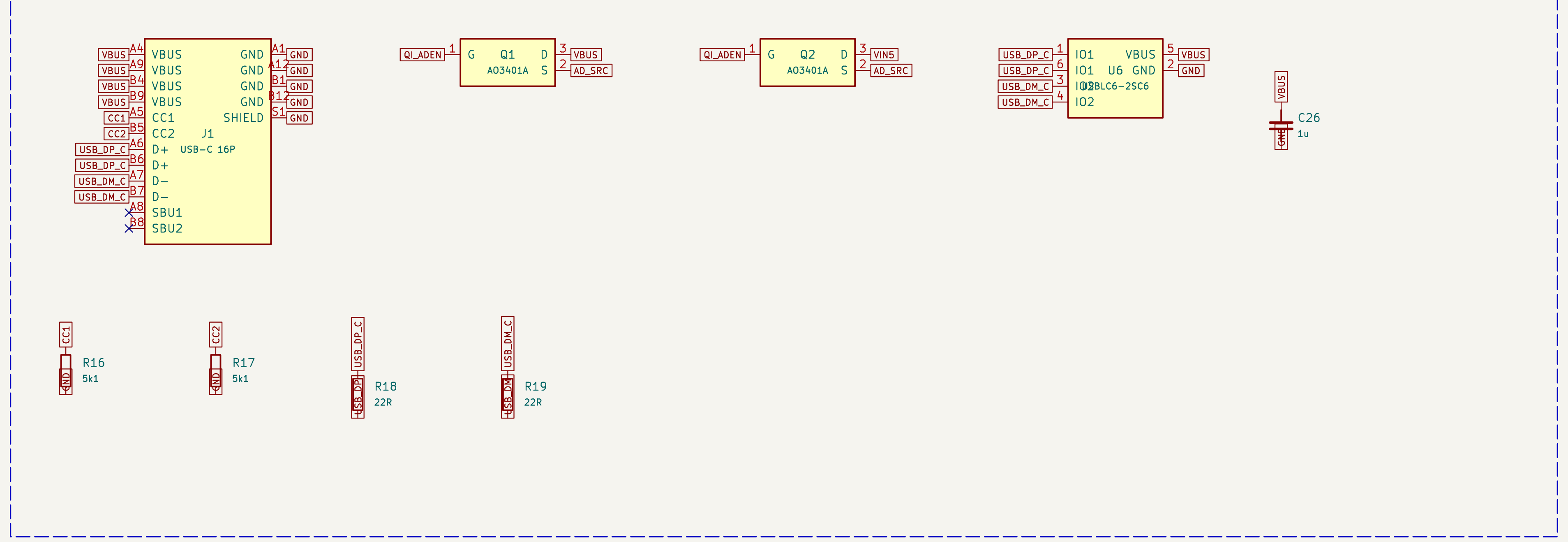
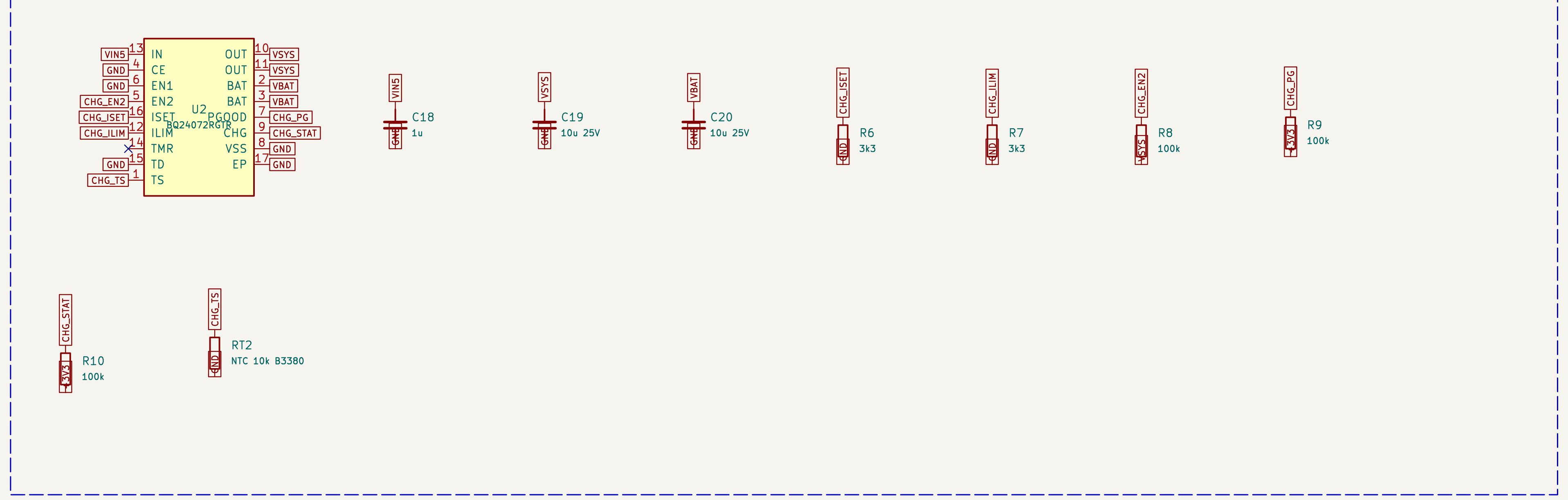
RECEPTOR Qi (BQ51013B) · bobina Ø25 · USB por AD/AD-EN


U1
BQ51013BRHLR
AC1
AC2
BOOT1
BOOT2
CLAMP1
CLAMP2
COMM1
COMM2
AD
AD-EN
EN1
EN2
RECT
OUT
CHG
ILIM
FOD
TS/CTRL
PGND
EP
QI_AC1
QI_AC2
QI_BOOT1
QI_BOOT2
QI_CLAMP1
QI_CLAMP2
QI_COMM1
QI_COMM2
VBUS
QI_ADEN
GND
QI_RECT
VIN5
QI_CHG
QI_ILIM
QI_FOD
QI_TS
C1 100n C0G 50V
C2 100n C0G 50V
C3 47n C0G 50V
C4 2n2 C0G 50V
C5 1n C0G 50V
C6 10n 50V
C7 470n 25V
C8 22n 50V
C9 10n 50V
C10 470n 25V
C11 22n 50V
C12 10u 25V
C13 10u 25V
C14 100n
C15 10u 25V
C16 100n
C17 1u
L1 BOBINA Qi (1)
L2 BOBINA Qi (2)
QI_COIL1
QI_AC2
R1 330R
R2 200R
R3 20k
R4 100k
RT1 NTC 10k B3380

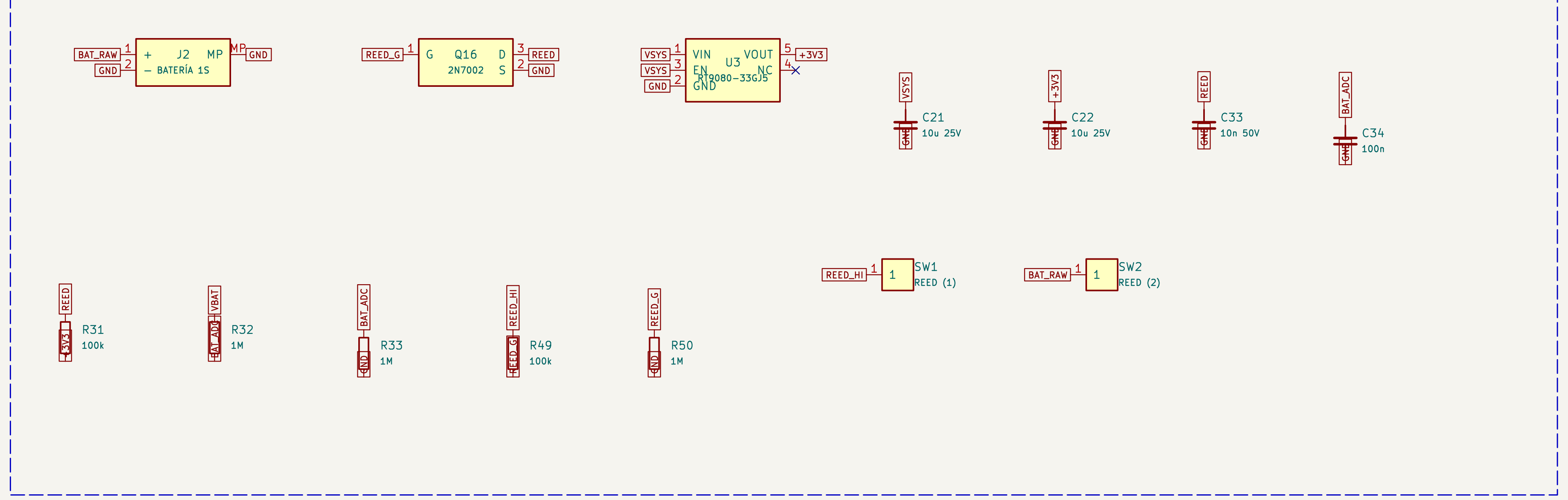
USB-C (programación) · ESD · entrada AD


J1
USB-C 16P
VBUS
CC1
CC2
D+
D-
SBU1
SBU2
GND
SHIELD
USB_DP_C
USB_DM_C
Q1 AO3401A
Q2 AO3401A
QI_ADEN
G
D
S
AD_SRC
VIN5
U6
USBLC6-2SC6
IO1
IO2
C26 1u
R16 5k1
R17 5k1
R18 22R
R19 22R

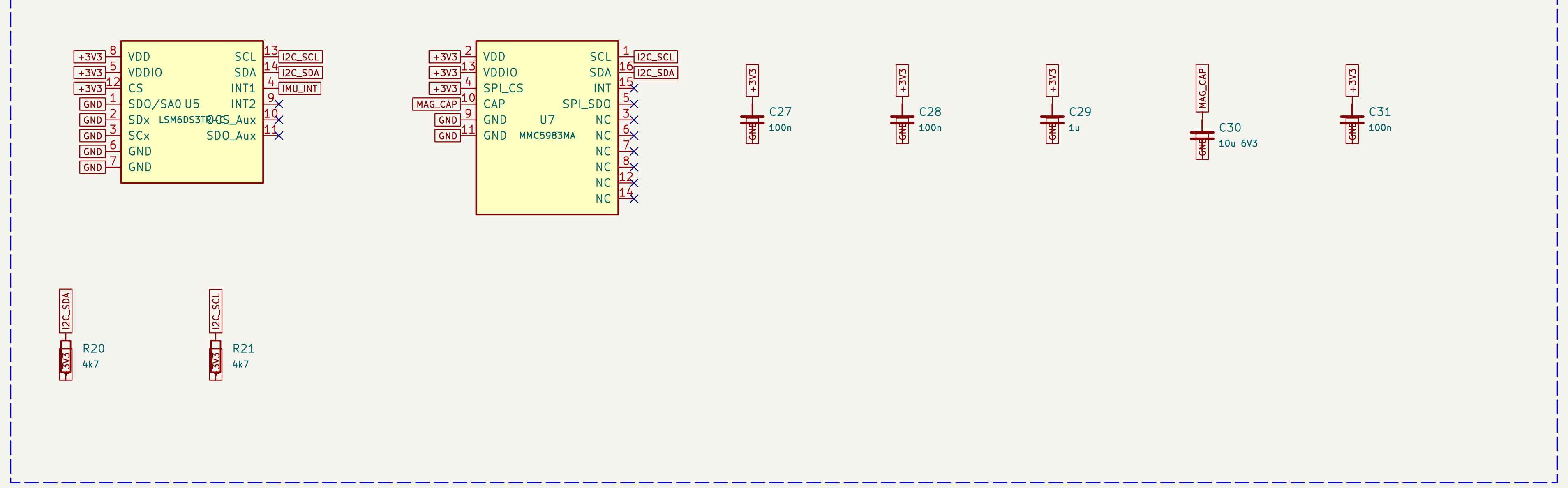
CARGADOR CON CAMINO DE POTENCIA (BQ24072) · batería 1S 500 mAh


U2
BQ24072RGTR
IN
CE
EN1
EN2
ISET
ILIM
TMR
TD
TS
OUT
BAT
PGOOD
CHG
VSS
EP
VIN5
VSYS
VBAT
CHG_PG
CHG_STAT
GND
C18 1u
C19 10u 25V
C20 10u 25V
R6 3k3
R7 3k3
R8 100k
R9 100k
R10 100k
RT2 NTC 10k B3380

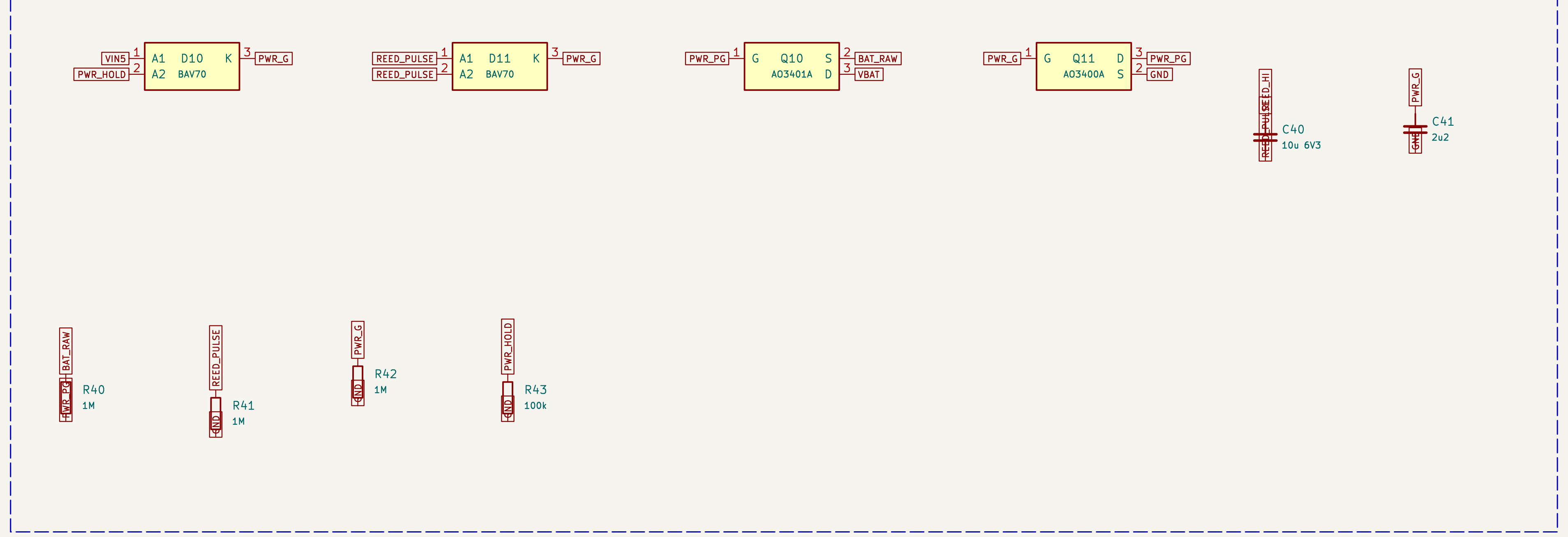
3,3 V (RT9080) · REED · BATERÍA (ADC)


J2
BATERÍA 1S
BAT_RAW
GND
Q16 2N7002
REED_G
REED
U3
RT9080-33GJ5
VIN
EN
GND
VOUT
NC
VSYS
+3V3
C21 10u 25V
C22 10u 25V
C33 10n 50V
C34 100n
SW1 REED (1)
SW2 REED (2)
REED_HI
R31 100k
R32 1M
R33 1M
R49 100k
R50 1M

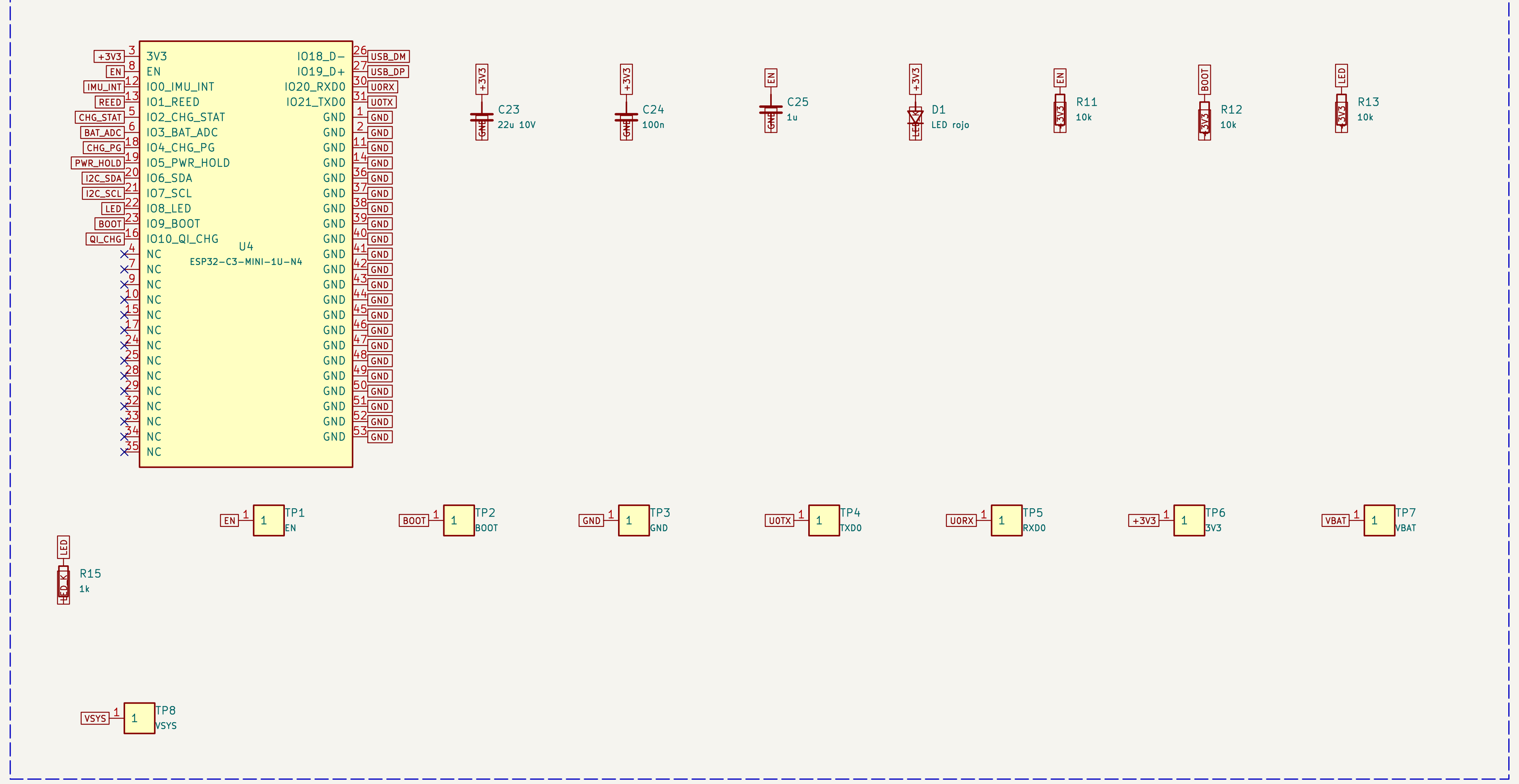
IMU LSM6DS3TR-C + MMC5983MA (I²C)


U5
LSM6DS3TR-C
VDD
VDDIO
CS
SDO/SA0
SDx
SCx
GND
SCL
SDA
INT1
INT2
OCS_Aux
SDO_Aux
U7
MMC5983MA
SPI_CS
CAP
SPI_SDO
NC
I2C_SCL
I2C_SDA
IMU_INT
MAG_CAP
+3V3
C27 100n
C28 100n
C29 1u
C30 10u 6V3
C31 100n
R20 4k7
R21 4k7

CORTE DE BATERÍA: enclavamiento con MOSFET (reed · 5 V · PWR_HOLD)


D10 BAV70
D11 BAV70
A1
A2
K
VIN5
PWR_HOLD
PWR_G
REED_PULSE
Q10 AO3401A
Q11 AO3400A
BAT_RAW
VBAT
PWR_PG
C40 10u 6V3
C41 2u2
R40 1M
R41 1M
R42 1M
R43 100k

ESP32-C3-MINI-1U-N4 · arranque · LED · pruebas


U4
ESP32-C3-MINI-1U-N4
3V3
EN
IO0_IMU_INT
IO1_REED
IO2_CHG_STAT
IO3_BAT_ADC
IO4_CHG_PG
IO5_PWR_HOLD
IO6_SDA
IO7_SCL
IO8_LED
IO9_BOOT
IO10_QI_CHG
NC
IO18_D-
IO19_D+
IO20_RXD0
IO21_TXD0
GND
+3V3
IMU_INT
REED
CHG_STAT
BAT_ADC
CHG_PG
PWR_HOLD
I2C_SDA
I2C_SCL
LED
BOOT
QI_CHG
USB_DM
USB_DP
U0RX
U0TX
C23 22u 10V
C24 100n
C25 1u
D1 LED rojo
R11 10k
R12 10k
R13 10k
R15 1k
TP1 EN
TP2 BOOT
TP3 GND
TP4 TXD0
TP5 RXD0
TP6 3V3
TP7 VBAT
TP8 VSYS

Generado por fuente/avo_girth_pcb.py (mismo origen de datos que la placa). Las conexiones son etiquetas globales: dos pines con la misma etiqueta están unidos.

AVO Performance Horse
Sheet: /
File: AVO_GIRTH_PCB.kicad_sch
Title: AVO GIRTH · placa de la cincha · REV B
Size: User
Date: 2026-10-29
Rev: B
KiCad E.D.A. 9.0.9
Id: 1/1